

Marketing Analytics

Chapter 2 · Describing Data

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Before we start

Paper quiz on Chapter 1 (*10 min*)

- Individual, closed book
- Counts toward participation
- Questions look like the learning check at the end of last session

Setup check

- Who completed the **R onboarding** checkpoint (report knitted to HTML)?
- Still stuck? → **setup clinic** now, during the quiz debrief — bring your laptop and the error message

Today's session

Part A — Getting to know the data

- 2.1 A first conversation with your data
- 2.2 Getting data into R
- 2.3 Variables and levels of measurement
- 2.4 Data handling

Part B — Describing it well

- 2.5 Not trusting your data too early
- 2.6 Summarizing data
- 2.7 Visualizing data
- 2.8 From description to insight

Laptops open: today is a live-coding session. Code along in your RStudio Project.

2.1 A first conversation with your data

Your first day in the data team

The head of A&R at a record label hands you a data set of tracks from the streaming charts:

“What is going on in the charts?”

- Which genres dominate?
- How concentrated is success?
- What distinguishes the hits from the rest?

Driving question of this chapter: *What is going on?* — the **descriptive** question from §1.3.

Before any model: **look at the data.**
Description is where the problems are caught.

Four data sets — one set of statistics

	Mean of x	Mean of y	SD of x	SD of y	Correlation
Data set 1	9	7.5	3.32	2.03	0.82
Data set 2	9	7.5	3.32	2.03	0.82
Data set 3	9	7.5	3.32	2.03	0.82
Data set 4	9	7.5	3.32	2.03	0.82

Same means, same standard deviations, same correlation. **Are these four data sets alike?**

Summaries compress — pictures reveal



A noisy line, a perfect curve, a line with one outlier, a “relationship” created by a single point — **identical summary statistics**. Good description always uses **both** numbers and pictures.

Anscombe (1973), *The American Statistician* — built-in R data set [anscombe](#).

The data: tracks in the streaming charts

~67,000 tracks that appeared in the Spotify charts, with 31 variables:

Group	Variables
Identifiers	isrc, artist_id, artistName, trackName
Chart performance	streams, weeks_in_charts, n_regions, top10
Audio characteristics (0–100)	danceability, energy, speechiness, instrumentalness, liveness, valence
Track properties	tempo, song_length, song_age, explicit, release_date, genre, label
Other platforms & reach	youtube_views, tiktok_counts, shazam_counts, n_playlists, artist-level followers & listeners
Expert judgment	expert_rating (poor – fair – good – excellent – masterpiece)

2.2 Getting data into R

Reading the data

```
1 library(tidyverse)
2
3 music_url <- "https://raw.githubusercontent.com/wu-rds/MA2026/main/data/music_data_fin.csv"
4 music_data <- read_csv2(music_url)
```

Why `read_csv2()` and not `read_csv()`? What do you expect to happen with `read_csv()`?

An import can fail silently

```
1 wrong_import <- read_csv(music_url)
2 dim(wrong_import)
```

```
[1] 66780      1
```

Convention	Columns separated by	Decimal mark	Function
International	,	.	<code>read_csv()</code>
European	;	,	<code>read_csv2()</code>

No error ≠ correct. Worse variants go unnoticed: `3,14` silently becoming `314`. Always inspect right after importing.

A first look

```
1 music_raw |>
2   select(streams, danceability, genre, label, release_date, expert_rating) |>
3   glimpse()
```

Rows: 66,796

Columns: 6

```
$ streams      <dbl> 11944813, 8934097, 38835, 46766, 2930573, 72199738, 1041694, 44363...
$ danceability <dbl> 50.9, 35.3, 68.3, 70.4, 84.2, 35.2, 73.0, 55.6, 71.9, 34.6, 82.0, ...
$ genre       <chr> "other", "Pop", "Pop", "HipHop/Rap", "HipHop/Rap", "Rock", "HipHop...
$ label       <chr> "Independent", "Universal Music", "Universal Music", "Independent"...
$ release_date <date> 2016-06-20, 2018-06-14, 2018-04-26, 2020-03-31, 2020-02-07, 2004-...
$ expert_rating <chr> "excellent", "good", "good", "poor", "excellent", "fair", "poor", ...
```

```
1 summary(select(music_raw, streams, song_length))
```

streams		song_length	
Min. :	1003	Min. :	0.01667
1st Qu.:	55760	1st Qu.:	2.92925
Median :	333336	Median :	3.33978
Mean :	7314674	Mean :	3.43116
3rd Qu.:	2181086	3rd Qu.:	3.81987
Max. :	2165692479	Max. :	29.78383

Other formats — same logic

Format	Package	Function
CSV (international / European)	readr	<code>read_csv()</code> / <code>read_csv2()</code>
Other delimiters	readr	<code>read_delim("file.txt", delim = "\t")</code>
Excel	readxl	<code>read_excel("file.xlsx", sheet = "Sheet1")</code>
SPSS / Stata	haven	<code>read_sav()</code> / <code>read_dta()</code>
Qualtrics export	qualtRics	<code>read_survey()</code> → group project
Web pages / APIs	rvest / jsonlite	see the course book (<i>optional</i>)

Files in your project folder are read by their **relative path**: `read_excel("data/survey.xlsx")` — no `setwd()` needed inside an RStudio Project.

What is a row?

Tidy data (Wickham, 2014):

1. Each **variable** is a column
2. Each **observation** is a row
3. Each **value** is a cell

Untidy: one column per month — “month” is hidden in the column names

track	Jan	Feb	Mar
Track A	120	135	150
Track B	80	70	65

Before any analysis, answer explicitly: **what does one row represent?** — the **unit of observation** (cf. “unit of analysis” in §1.8). Here: one row = one track. *Does that really hold?* → \$2.5

Tidy: one row per track and month

track	month	streams
Track A	Jan	120
Track A	Feb	135
Track A	Mar	150
Track B	Jan	80
Track B	Feb	70
Track B	Mar	65

2.3 Variables and levels of measurement

Levels of measurement: categorical

Nominal

- Numbers only serve as **labels** for identification and categorization
- They do not reflect the **amount** of a characteristic
- Two categories: **binary**
- Permissible operation: **counting**

E.g., starting numbers in a race



$7 \cdot 11 \cdot 3$

Ordinal

- Numbers indicate the **relative position** (order)
- ...but not the **size of the differences**
- Also: **greater / less than**; percentile-based statistics such as **median** and **quartiles**

E.g., finishing order in the race



1st · 2nd · 3rd place

Levels of measurement: continuous (metric)

Interval

- **Differences** can be compared (equal intervals)
- ...but the **zero point is arbitrary**
- Ratios are **not** meaningful
- Also: range, mean, standard deviation

E.g., finishing clock time (start: 14:00)



14:07:06 · 14:14:12 · 14:15:12

Ratio

- All properties of the scales above
- **Absolute zero point**
- Ratios are meaningful (“**twice as fast**”)
- All statistical techniques apply

E.g., time to finish (minutes)



7.1 · 14.2 · 15.2

Not every number is a quantity

Level	Values express	Music data	Meaningful	Summary
Nominal	Categories, no order	genre, label, artist_id	= / ≠	Frequencies, mode
Ordinal	Ordered categories	expert_rating	+ < / >	+ median, quantiles
Interval	Equal distances, arbitrary zero	release_year, °C	+ differences	+ mean, SD
Ratio	Equal distances, natural zero	streams, song_length	+ ratios	+ coefficient of variation

Nominal + ordinal = **categorical**; interval + ratio = **continuous**. The level decides which **summaries, charts** — and later **tests** — are meaningful. The mean `artist_id` is as meaningless as the mean phone number.

Your turn: classify these (2 min)

Nominal, ordinal, interval, or ratio?

- a. Customer ID · (b) Satisfaction: “very dissatisfied” ... “very satisfied” · (c) Price in € · (d) Payment method · (e) Year of first purchase · (f) Purchased (yes / no)

Telling R the level of measurement

```

1 music_data <- music_raw |>
2   mutate(
3     genre      = as.factor(genre),                # nominal
4     explicit   = factor(explicit, levels = c(0, 1),
5                          labels = c("not explicit", "explicit")), # nominal (binary)
6     artist_id  = as.character(artist_id),          # an ID is not a number
7     expert_rating = factor(expert_rating,
8                            levels = c("poor", "fair", "good", "excellent", "masterpiece"),
9                            ordered = TRUE)          # ordinal
10  )

```

```
1 table(music_data$expert_rating)
```

poor	fair	good	excellent	masterpiece
5356	13418	23243	23376	1403

As text, R would sort the ratings **alphabetically**; as an **ordered factor**, R knows that *good* lies between *fair* and *excellent*.

2.4 Data handling

The verbs of data handling

Verb	Does	Example question
<code>filter()</code>	keeps rows meeting a condition	Pop tracks with > 10M streams?
<code>select()</code>	keeps columns	only artist, title, streams
<code>arrange()</code>	sorts rows	the most-streamed tracks?
<code>mutate()</code>	creates / changes columns	streams in millions? release year?
<code>group_by() + summarise()</code>	condenses to one row per group	median streams per label?
<code>count()</code>	counts rows per group	tracks per genre?
<code>left_join()</code>	combines tables via a key	add label information

Same logic everywhere: data frame in → **column names without quotes** → data frame out. Chained with the pipe `|>`: “*and then*”.

Chaining steps with the pipe

```

1 music_data |>                                # take the music data, then
2   filter(genre == "Pop") |>                  # keep the pop tracks, then
3   arrange(desc(streams)) |>                 # sort by streams, then
4   select(artist, track, streams) |>         # keep three columns, then
5   head(4)                                   # show the first four rows

```

A tibble: 4 × 3

	artist	track	streams
	<chr>	<chr>	<dbl>
1	Ed Sheeran	Shape of You	2165692479
2	Shawn Mendes feat. Camila Cabello	Señorita	1156206588
3	XXXTENTACION	SAD!	1103693478
4	Ed Sheeran	Perfect	1045189446

“... per ...”: group_by() + summarise()

```
1 music_data |>
2   group_by(label) |>
3   summarise(
4     n_tracks      = n(),
5     median_streams = median(streams),
6     share_top10 = scales::percent(mean(top10 == "yes"), accuracy = 0.1)
7   )
```

A tibble: 4 × 4

	label	n_tracks	median_streams	share_top10
	<fct>	<int>	<dbl>	<chr>
1	Independent	22570	238884	7.9%
2	Sony Music	12390	355898.	11.3%
3	Universal Music	21632	434943	11.3%
4	Warner Music	10204	347152.	10.2%

The workhorse of descriptive analysis: every question of the form “... per ...” has this shape.

Combining tables: left_join()

```
1 label_info <- tibble(  
2   label      = c("Universal Music", "Sony Music", "Warner Music", "Independent"),  
3   label_type = c("Major", "Major", "Major", "Independent")  
4 )  
5 music_data <- music_data |> left_join(label_info, by = "label")  
6 music_data |> count(label_type)
```

A tibble: 2 × 2

	label_type	n
	<chr>	<int>
1	Independent	22570
2	Major	44226

Keeps all rows of the left table; no match → **NA**. Check the row count after every join — a non-unique key duplicates rows silently.

Your turn (live coding, 8 min)

Using the verbs, answer:

1. How many **hip-hop** tracks reached the **top 10**?
2. Which **label** has the highest **median streams** among hip-hop tracks?
3. Create a variable `streams_mio` (streams in millions) and show the five **rock** tracks with the most streams.

2.5 Not trusting your data too early

Five checks before you trust a data set

1. **Did the import work?** Rows, columns, types — §2.2
2. **Missing values** — including zeros that mean “unknown”
3. **Impossible values** — ranges, dates, units
4. **Duplicates** — is the identifier really unique?
5. **Extreme values** — errors or information?

...and one question beyond the checks: **how did the data come to be?**

Check 2: zeros that mean “unknown”

```
1 sum(is.na(music_data))           # missing values: none at all?
```

```
[1] 0
```

```
1 mean(music_data$youtube_views == 0)   # share of tracks with 0 YouTube views
```

```
[1] 0.5796155
```

58% of charting tracks with **zero** YouTube views? Almost certainly “**no data**”, not “no views”. **Disguised missing values** (revenue 0 for never-contacted customers, age 0 in the CRM) drag every average down. Recode: `na_if(youtube_views, 0)` — and report how many are missing.

Check 3: impossible values

```
1 music_data |>
2   summarise(first_release = min(release_date), last_release = max(release_date),
3             min_length = min(song_length), max_length = max(song_length),
4             min_tempo = min(tempo))
```

A tibble: 1 × 5

	first_release	last_release	min_length	max_length	min_tempo
	<date>	<date>	<dbl>	<dbl>	<dbl>
1	1942-01-01	2029-04-06	0.0167	29.8	0

Releases in **2029**, a **1-second** song, a tempo of **0 bpm**. Some values are clearly errors, others merely unusual (intros, medleys) — **inspect individually** before deciding.

Check 4: what is a row, really?

```
1 music_data |> summarise(n_rows = n(), n_distinct_isrc = n_distinct(isrc))
```

```
# A tibble: 1 × 2  
  n_rows n_distinct_isrc  
  <int>      <int>  
1  66796         56945
```

The ISRC should identify a recording uniquely — yet there are **9,851 more rows than ISRCs**, and the rows differ in streams and chart weeks. Versions? Periods? Errors? Without documentation we cannot tell — but “one row = one recording” **does not hold**.

Check 5: outliers are not always errors

```
1 music_data |> arrange(desc(streams)) |> select(artist, track, streams) |> head(4)
```

```
# A tibble: 4 × 3
```

	artist	track	streams
	<chr>	<chr>	<dbl>
1	Ed Sheeran	Shape of You	2165692479
2	Tones and I	Dance Monkey	1909947000
3	Billie Eilish	bad guy	1459149566
4	Lewis Capaldi	Someone You Loved	1419867299

In markets for creative products, **a few hits account for a large share of consumption**. Here, the extremes are the story — check them, don't delete them.

Document, don't silently delete: corrections in code, explicit flags instead of overwriting, transparent reporting of what changed.

How did the data come to be?



Only **charted** tracks are in the data. Catchiness and promotion are unrelated among all songs (left) — but **negatively related among the survivors** (right). Created purely by the selection.

Survivorship is everywhere in marketing data

- Successful **products** only (launch data of products still on the market)
- **Customers who did not churn** (loyalty-program data)
- **Companies that still exist** (“what do successful firms do?”)
- **Survey respondents** — who chooses to answer is itself a choice (→ your group project!)

Add one question to every descriptive checklist: **who is *not* in my data — and why?** Relationships among the survivors can differ from those in the population you care about (*Berkson's paradox*).

2.6 Summarizing data

Categorical variables: frequencies

```
1 music_data |>
2   count(genre, sort = TRUE) |>
3   mutate(share = n / sum(n)) |>
4   head(5)
```

A tibble: 5 × 3

	genre <fct>	n <int>	share <dbl>
1	Pop	30069	0.450
2	HipHop/Rap	21131	0.316
3	other	5228	0.0783
4	Rock	4214	0.0631
5	Electro/Dance	2703	0.0405

Conditional frequencies

```
1 music_data |>
2   group_by(genre) |>
3   summarise(n = n(), share_top10 = mean(top10 == "yes")) |>
4   arrange(desc(share_top10)) |>
5   mutate(share_top10 = scales::percent(share_top10, accuracy = 0.1))
```

A tibble: 11 × 3

	genre	n	share_top10
	<fct>	<int>	<chr>
1	Country	504	26.4%
2	other	5228	19.4%
3	R&B	1881	17.2%
4	Electro/Dance	2703	14.2%
5	Soundtrack	326	13.8%
6	Reggae	121	11.6%
7	HipHop/Rap	21131	9.7%
8	Pop	30069	7.9%
9	Rock	4214	7.4%
10	German Folk	539	5.2%
11	Classics/Jazz	80	1.2%

Beware of **small groups**: a share computed from 80 tracks is far less reliable than one from 30,000 — made precise in Module 3.

Mean or median?

```
1 music_data |>
2   summarise(mean = mean(streams), median = median(streams),
3             q25 = quantile(streams, 0.25), q75 = quantile(streams, 0.75))
```

A tibble: 1 × 4

	mean	median	q25	q75
	<dbl>	<dbl>	<dbl>	<dbl>
1	7314674.	333336.	55760	2181086.

“The average track has **7.3 million** streams.” — Is this a good description of the typical track?

No: a few mega-hits pull the mean up — the distribution is **right-skewed**. The **median** (~333,000) describes the typical track. Mean far above median → a few extremes dominate the average (sales, revenue, followers, streams).

Location, dispersion, shape — in one call

```
1 library(psych)
2 music_data |>
3   select(streams, danceability, energy, valence, song_length) |>
4   describe() |>
5   select(n, mean, sd, median, min, max, skew)
```

	n	mean	sd	median	min	max	skew
streams	66796	7314673.94	39956263.68	333335.50	1003.00	2165692479.00	16.74
danceability	66796	66.32	14.71	68.00	0.00	98.30	-0.53
energy	66796	64.37	17.52	66.10	0.00	100.00	-0.54
valence	66796	50.42	22.26	50.00	0.00	99.00	0.08
song_length	66796	3.43	0.86	3.34	0.02	29.78	3.36

- **Location:** mean, median
- **Dispersion:** SD, range, IQR, coefficient of variation (SD / mean)
- **Shape:** skewness — `streams` ≈ 17 : extremely right-skewed

z-scores express relative position: $z = (x - \bar{x})/s \rightarrow$ “1.5 SDs above the average track”. Their link to probabilities: Module 3.

2.7 Visualizing data

The grammar of graphics

Every `ggplot2` chart is built from the same components, **layer** by **layer** with `+`:

1. **Data** — `ggplot(music_data, ...)`
2. **Aesthetic mappings** — `aes(x = ..., y = ..., color = ...)`
3. **Geoms** — `geom_col()`, `geom_point()`, `geom_boxplot()`, ...
4. **Labels, scales, themes** — `labs()`, `scale_...()`, `theme_...()`

```
1 genre_shares <- music_data |>
2   count(genre) |> mutate(share = n / sum(n))
3 ggplot(genre_shares, aes(x = genre, y = share)) +
4   geom_col()
```



From default to useful

```

1 ggplot(genre_shares,
2       aes(x = fct_reorder(genre, share), y = share)) +
3   geom_col(fill = "#0f5da8") +
4   geom_text(aes(label = scales::percent(share, accuracy = 0.1)),
5             hjust = -0.15, size = 3.5) +
6   coord_flip() +
7   scale_y_continuous(labels = scales::percent,
8                     limits = c(0, 0.5)) +
9   labs(x = NULL, y = "Share of tracks",
10        title = "Pop and hip-hop dominate the charts") +
11   theme_minimal()

```



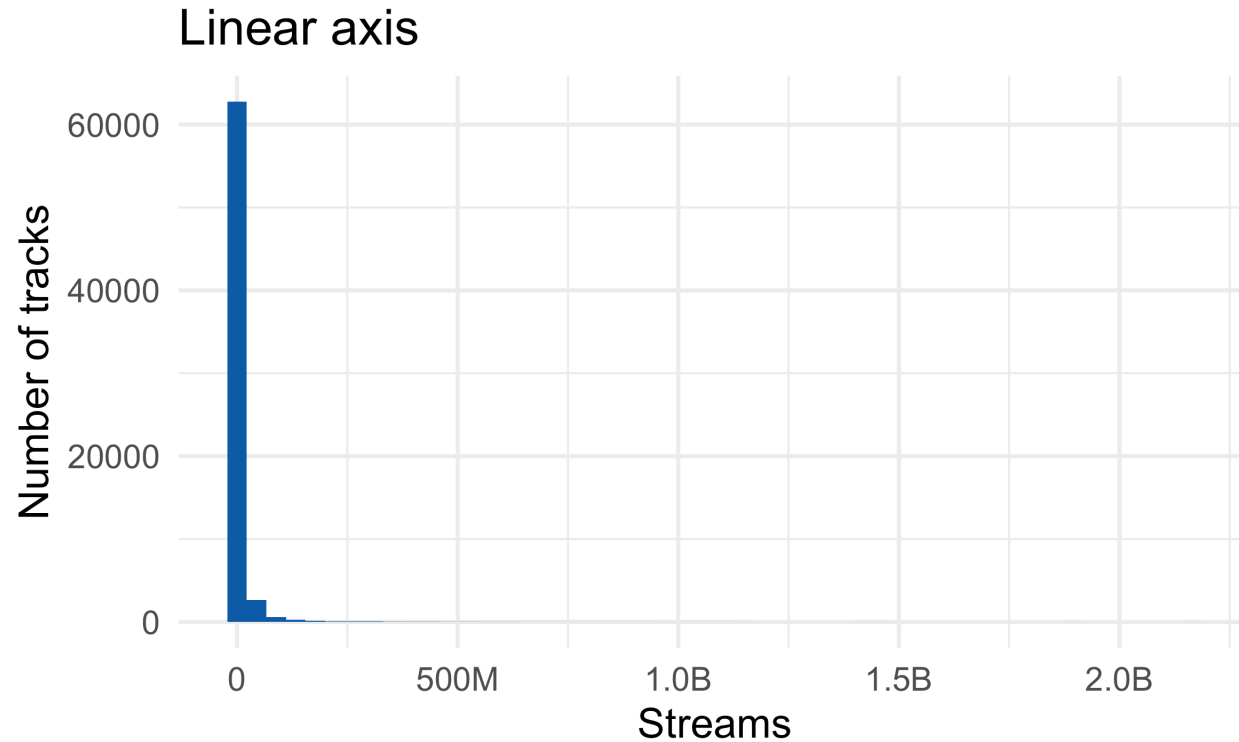
Sorted bars · readable labels · percent axis · a title that **states the finding**, not the topic.

Which chart for which variables?

To show ...	Variables	Chart	geom
One distribution	categorical	Bar chart	<code>geom_bar() / geom_col()</code>
	continuous	Histogram, density, boxplot	<code>geom_histogram(), geom_density(), geom_boxplot()</code>
Groups compared	categorical × continuous	Boxplots, jittered points	<code>geom_boxplot(), geom_jitter()</code>
A relationship	continuous × continuous	Scatter plot	<code>geom_point()</code>
Two categoricals	categorical × categorical	Grouped / stacked bars of shares	<code>geom_col(position = "dodge")</code>
Change over time	time × continuous	Line chart	<code>geom_line()</code>

A third variable: **color** (`aes(color = ...)`) or **small multiples** (`facet_wrap(~ ...)`).

Skewed data: the log scale



On a log scale, each step **multiplies** (10K → 100K → 1M). Makes skewed distributions readable — **always label it** in the axis title.

Comparing groups: boxplots



Charts are data-quality tools, too



Two things in this chart should make you suspicious. **Which?**

Honest visualization I: bars start at zero



Bars encode values by **length** — a truncated axis makes the lengths lie. (Points and lines encode by *position* and may start elsewhere — clearly labeled.)

Honest visualization II: dual axes



Remember this one from §1.4?

Two independently chosen y-axes let you make **almost any two series look related**.
Alternatives: index both series (first year = 100), or two charts side by side.

- Also avoid: **3D effects**, **pies** with many slices, colors without meaning, unsorted categories, axes without units

2.8 From description to insight

“Do explicit tracks perform worse?”

```
1 music_data |>
2   group_by(explicit) |>
3   summarise(n = n(), median_streams = median(streams))
```

```
# A tibble: 2 × 3
  explicit      n median_streams
  <fct>      <int>          <dbl>
1 not explicit 58603      385290
2 explicit    8193       93600
```

Can we write “*explicit lyrics reduce streams*”? What would you check first?

Looking within genres

```
1 music_data |>
2   mutate(genre = fct_lump_n(genre, n = 5, other_level = "Remaining genres")) |>
3   group_by(genre) |>
4   summarise(share_explicit      = mean(explicit == "explicit"),
5             median_not_explicit = median(streams[explicit == "not explicit"]),
6             median_explicit    = median(streams[explicit == "explicit"]))
```

A tibble: 6 × 4

	genre <fct>	share_explicit <dbl>	median_not_explicit <dbl>	median_explicit <dbl>
1	Electro/Dance	0.340	494986.	72608
2	HipHop/Rap	0.0515	618344.	226571
3	other	0.0779	704424	133351
4	Pop	0.155	236848.	87798
5	Rock	0.172	159262.	31128
6	Remaining genres	0.112	536213	224444

The gap persists within each large genre — but its size **and the share of explicit tracks** vary strongly. Genre, label, artist popularity, and the selection into the charts are all candidate **confounders** (§1.5).

What we can — and cannot — write

Descriptive ✓

“Among the tracks in our chart data, those with explicit lyrics have a lower median number of streams than those without (about 94,000 vs. 385,000). The gap is visible within each large genre, but varies in size.”

Causal ✗

“Explicit lyrics reduce streams.”

That would require comparing the **same** tracks with and without explicit lyrics — a counterfactual the data cannot provide (§1.2).

A good descriptive finding answers: **What** was measured? **How much** (numbers, not adjectives)? **Compared to what? For whom? With which caveat?** — and uses descriptive verbs: *is higher, differs, is associated with* — not *causes, drives, increases*.

Assignment 1: data handling & description

- **Your personal version of the data** (based on your student ID) — check, handle, summarize, visualize
- Submit as a **reproducible report** (.Rmd → HTML) via Canvas
- **Deadline:** Group A: Mon, Oct 26 · Group B: Wed, Oct 28
- **Debrief:** 3 short paper questions about *your* results at the start of session 4 (closed book)
- **10 points:** report 7 + debrief 3

Everything you need is in **this chapter**. AI assistants welcome — you must be able to **explain every line** you submit.

Learning check

1. After `read_csv("file.csv")` on a file from a German Excel, the data frame has **one column**. Why — and what should you use instead?
2. Mean revenue per customer: **€480**, median: **€95**. What does this tell you about the distribution? Which number describes the typical customer?
3. A loyalty-program data set contains **only active customers**. Among them, those who received few discounts buy most often. Why might this pattern mislead?
4. **True or false?** (a) A variable stored as a number can be nominal. (b) Outliers should always be removed. (c) The y-axis of a bar chart should start at zero. (d) A zero in a data set always means the quantity was zero.

Next session

Module 3 · Uncertainty & Experiments I — *How sure can I be?*

- Why every number from a sample is uncertain — and how to quantify it
- Standard errors and confidence intervals — by simulation
- Our first experiment analysis: comparing two groups

Course book chapter 2 (consolidation reading for today) is online — with all code from these slides. **Assignment 1** is on Canvas.